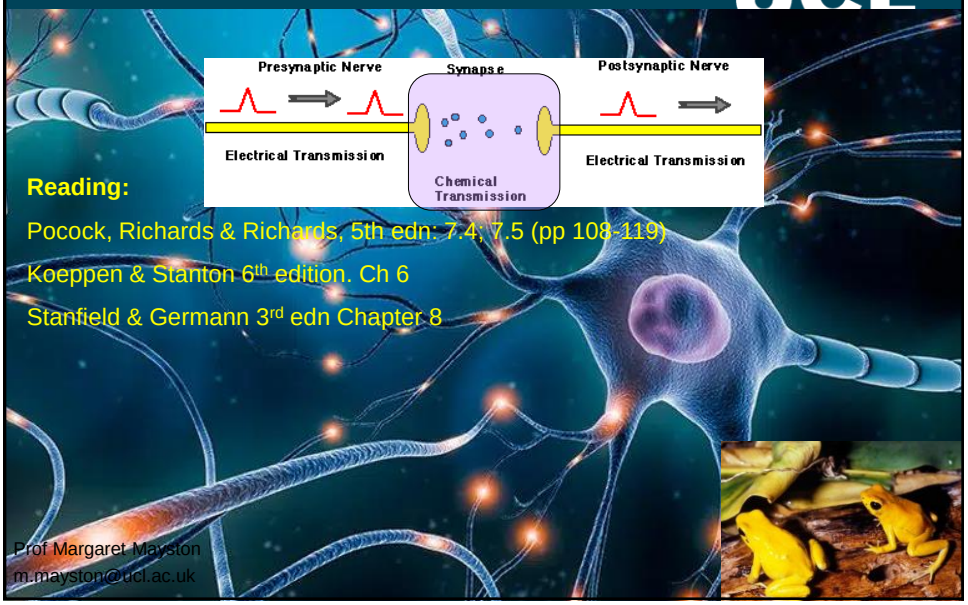
Graded potentials: Synaptic transmission



Reading:

Pocock, Richards & Richards, 5th edn: 7.4, 7.5 (pp 108-119)

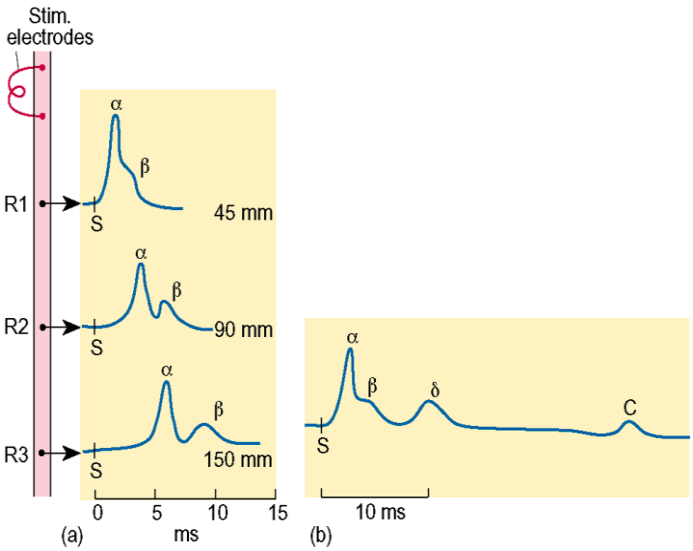
Koeppen & Stanton 6th edition. Ch 6

Stanfield & Germann 3rd edn Chapter 8

Prof Margaret Mayston
m.mayston@ucl.ac.uk

Action potential: Mechanisms of conduction

Compound action potentials show that some axons conduct faster than others



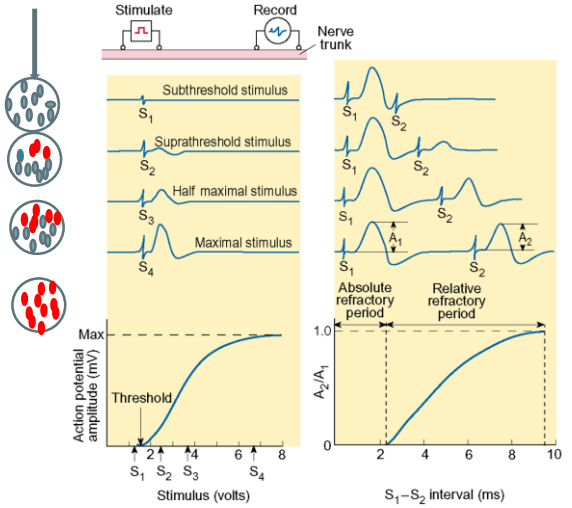
'A' fibres conduct at 30-70m/s depending on type (sensory afferents are fastest).

'C' fibres are the slowest: 0.5 - 2.0m/s. These are unmyelinated

2

Action potential: Mechanisms of conduction

Action potentials can be recorded ***from intact nerves***: **compound action potentials (CAP) when record from several axons**



2 shocks of equal intensity

The CAP represents the result of stimulation of several axons, hence its change in amplitude with increasing stimulus strength.

There is also a refractory period for whole nerve stimulation.

3

Fast and slow spiking neurons……

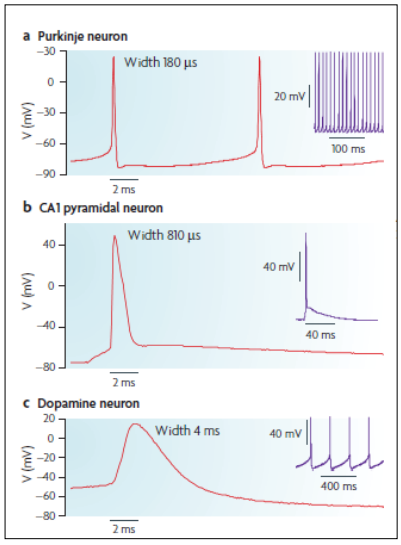
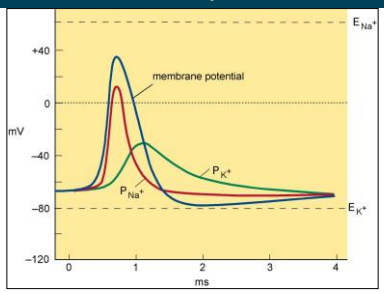


Fig.1 Bean, 2007

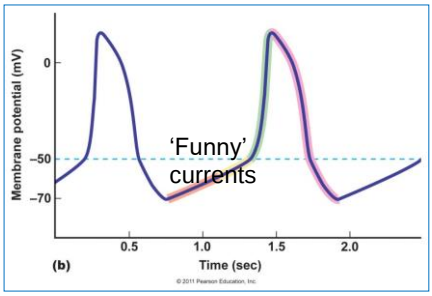
NATURE REVIEWS | NEUROSCIENCE

4

Neuronal action potential

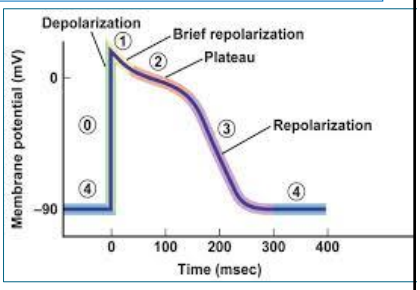


Cardiac action potentials



4 key points:
Threshold, time course and duration of the AP depend on:

1. The gating and permeability of specific types of ion channels- these depend on Vm and time.
2. The intracellular and extracellular concentrations of ions that pass through those channels: Na+, K+, Ca+ Cl-
3. Membrane properties such as capacitance, resistance and geometry of the cell
4. The shape of the AP reflects the specialized function of that cell.



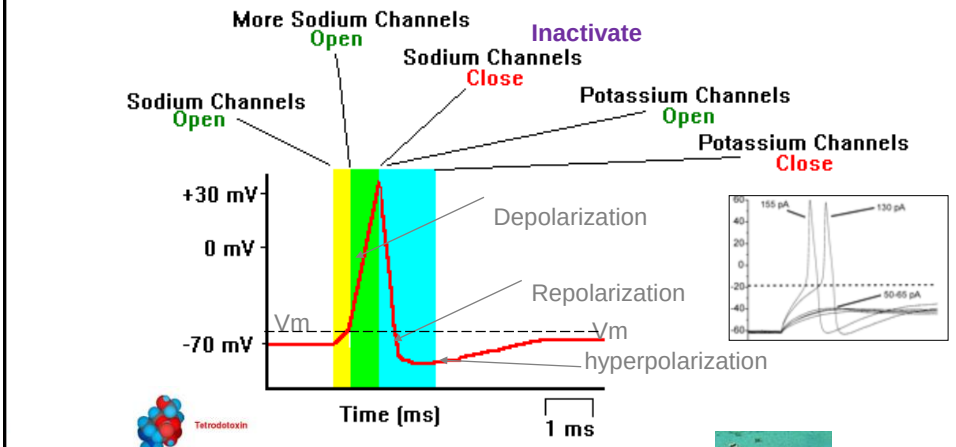
5

Learning outcomes:

At the end of this lecture you should be able to:

- Understand and explain synaptic transmission
- Understand excitatory & inhibitory synaptic transmission in the CNS
- Describe the process of excitatory & inhibitory synaptic transmission in the CNS
- Describe why and how synaptic potentials are graded
- Explain summation
- Understand the steps which take place during synaptic transmission at the neuromuscular junction
- Understand the relevance of synaptic activity for learning, memory and as a target for pharmacological intervention

Action potential: Sequence of opening and closing of voltage gated sodium and potassium channels- 'all or none':



Puffer fish TTX blocks Na^+ channels



7

Action potential vs Graded (synaptic)potential

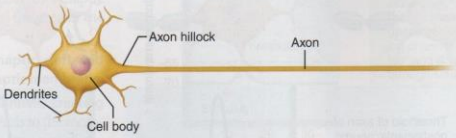
Property	Action potential	Synaptic (graded) potential
Amplitude	Large (70-110mV)	Small, but can summate to result in AP
Duration	Brief: usually 1-2 msec	Variable: 5msec to minutes
Conduction	All or none	Decremental
Summation	None, amplitude is same regardless of stimulus initiating it	Graded: depends on condition initiating it
Signal	Depolarizing	Depolarizing or hyperpolarizing
Refractory period	Absolute and relative	None
Distance	Long	Short

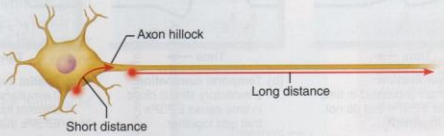
8

Graded potentials vs action potentials

Table 11.2 Comparison of Graded Potentials and Action Potentials

	GRADED POTENTIAL (GP)	ACTION POTENTIAL (AP)
Location of event	Cell body and dendrites, typically	Axon hillock and axon
Distance traveled	Short distance—typically within cell body to axon hillock (0.1–1.0 mm)	Long distance—from trigger zone at axon hillock through entire length of axon (a few mm to over a meter)



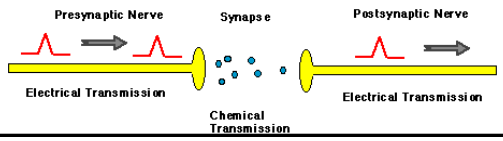


9

Synaptic transmission- key points



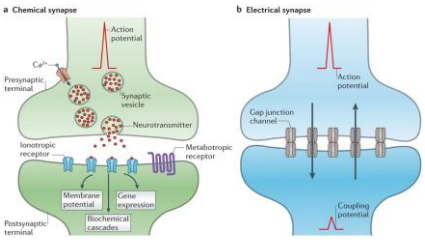
- A contact between 2 neurons is called a synapse (Sherrington)
- Synapses are the way that nerve cells/neurons can communicate with each other and thus the way in which different parts of the CNS can communicate and influence each other. i.e. the way that nerve cells signal their targets.
- Synaptic junctions can be located on different parts of the neuron
- Synaptic signals act on specific target cells e.g. neurons, muscle, glands
- Synapses can be of different types: excitatory (depolarising), inhibitory (hyperpolarising)
- Synaptic potentials are an example of graded potentials and can summate
- Synapses can be chemical or electrical (gap junctions); can be fast or slow
- Changes in synaptic efficiency are the basis of learning and memory



10

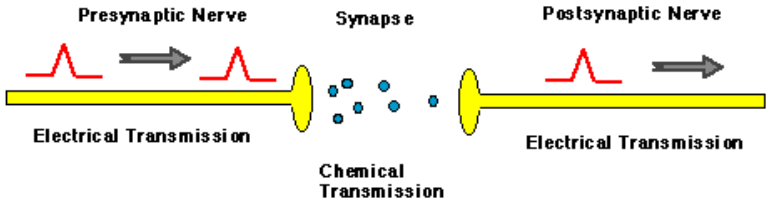
Different types of synapses

- Chemical: most usual type, associated with the secretion of a small amount of chemical (neurotransmitter) from the nerve terminal, and is a one-way signalling mechanism. These are complex because the pre- and post-synaptic membranes are separated by a minute space called the synaptic cleft, hence the need for a chemical neurotransmitter from the pre-synaptic fibre.
- Electrical: the electrical current generated by the AP is transmitted directly to the post synaptic cell by electrical coupling because the two cells are in tight contact with each other (gap junctions). This is fast and reliable. This type of transmission occurs in some retinal cells and in cardiac cells. Also occurs between some glial cells e.g. astrocytes.



11

Synaptic transmission [chemical] - key points



AP **cannot** cross the synaptic cleft: the synapse provides the solution.

End of nerve enlarges into axon terminal (synaptic bouton) containing vesicles which store transmitter

AP opens Ca^{2+} channels which cause some of vesicles to fuse to the membrane and they open and release their neurotransmitter

Transmitter travels across by simple diffusion- short distance

It binds to specific receptor protein in the post-synaptic membrane and ligand-gated channels are opened

If sufficient Excitatory Post Synaptic Potentials (EPSPs) to depolarise the membrane an AP will be generated and the signal will travel along

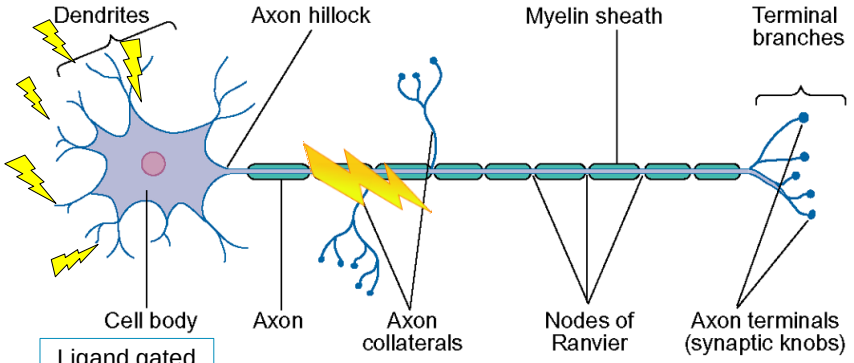
12

UCL

Main structural features of neurons

Dendrites = input

Integration



Ligand gated channels

voltage gated channels

Axons = communicate / output

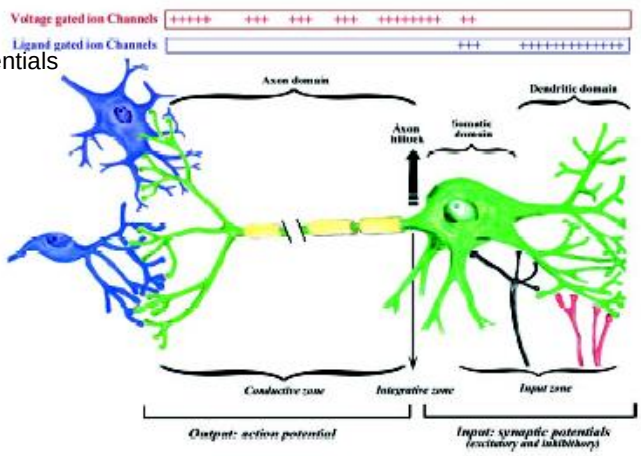
13

UCL

Signal transmission in neurons: ***structure = function***

Action potentials

Graded potentials



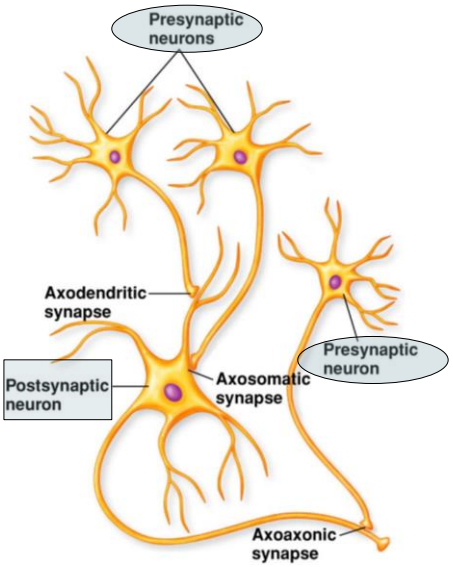
Giuliodori & DiCarlo, 2004

14

Types of synaptic contact between neurons

Neuron-to-Neuron Synapses

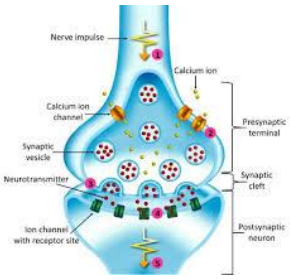
A synapse is a specialised junction where transmission of information takes place between a nerve fibre and another nerve cell (or muscle or gland). There are 3 basic sites for synapses.



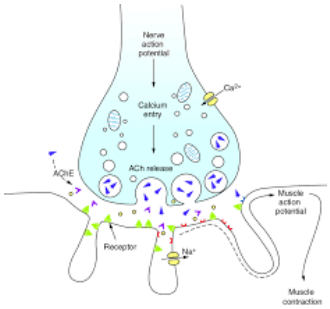
S&G Figure 8.1

15

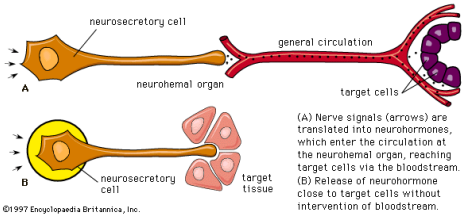
Neurons target specific cells: e.g. another neuron/neuromuscular junction/organ



Neuron to neuron



Neuromuscular junction



Target cell e.g. neurosecretory cell e.g. oxytocin via the blood stream

16

Detailed structure of a CNS chemical synapse

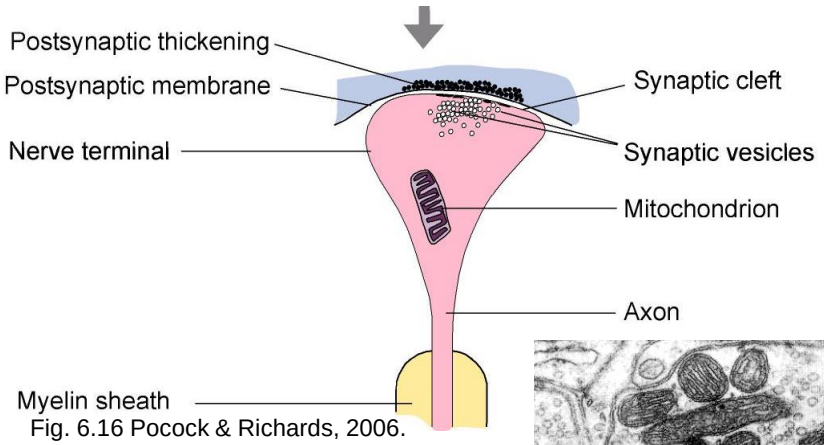


Fig. 6.16 Pocock & Richards, 2006.

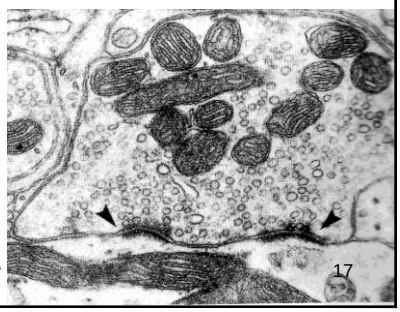
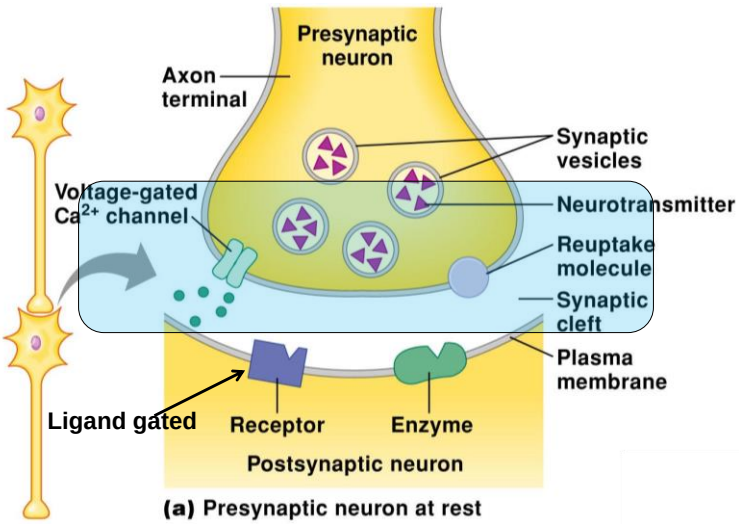


Fig. 10.6 Kandel, Schwartz & Jessell, 2000.

17



Requires transmitter specific receptor on post-synaptic membrane…….

18

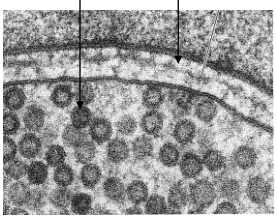
‘Docking’ and ‘priming’

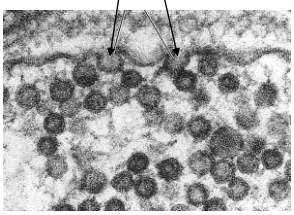
Fig. 14.8 Kandel, Schwartz & Jessell, 2000.

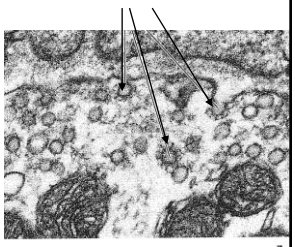
Vesicles & synaptic cleft

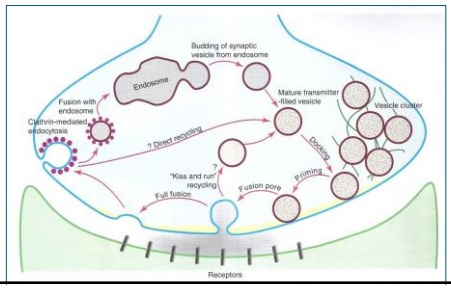
Vesicle fusions

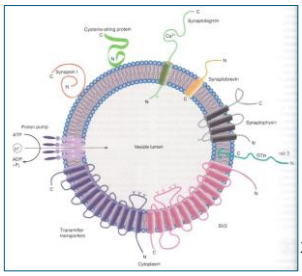
Coated vesicles





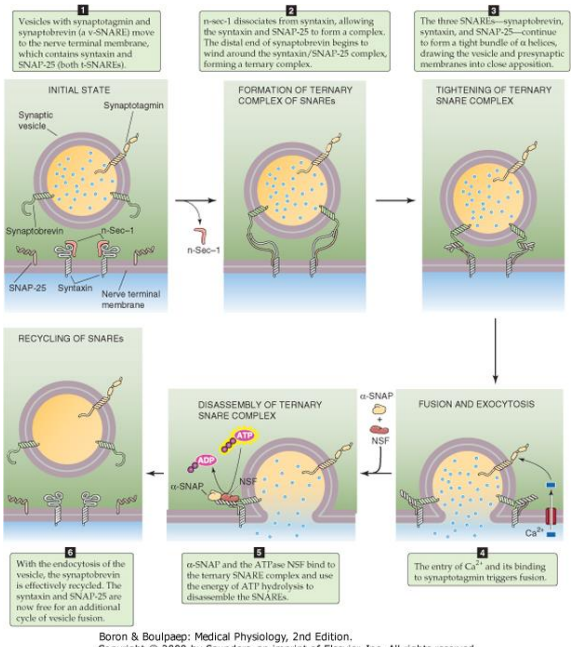






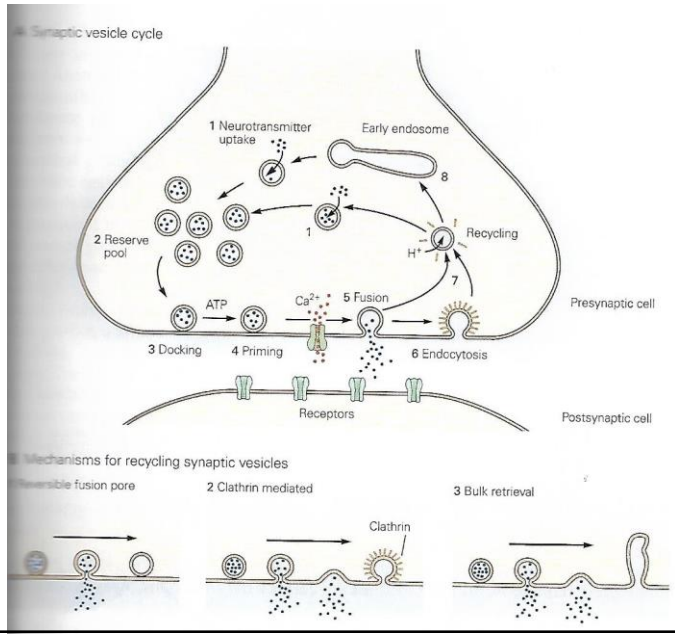
19

Vesicle membrane fusion and release:



20

Boron & Boulpaep: Medical Physiology, 2nd Edition.
Copyright © 2009 by Saunders, an imprint of Elsevier, Inc. All rights reserved.

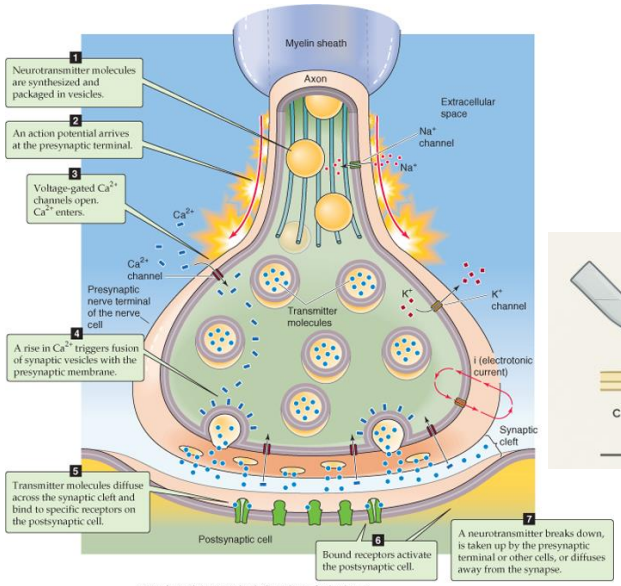


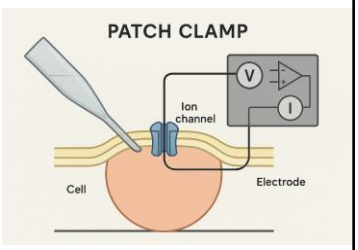
21

The main stages of synaptic transmission

1. **Action potential in axon of presynaptic neuron and depolarisation of nerve terminal**
2. **Opening of voltage-gated Ca^{2+} channels in terminal membrane**
3. **Calcium enters cell:**
4. **Secretion of neurotransmitter from vesicles in presynaptic membrane (released in packages: quanta) by exocytosis i.e. some of vesicles fuse to the presynaptic membrane and the vesicles open and release transmitter. (Ca^{2+} dependent)**
5. **Diffusion of transmitter across synaptic cleft.**
6. **Some of it binds to post-synaptic receptors (the protein structure of the receptor and the neurotransmitter are matched).**
7. **Response in post-synaptic cell: change of membrane potential in post-synaptic neuron, either depolarization or hyperpolarization.**
8. **Removal of neurotransmitter from synapse: i. some degraded by enzymes, ii. some taken up by pre-synaptic cell (endocytosis); iii. Some diffuses away from the synaptic cleft**

22

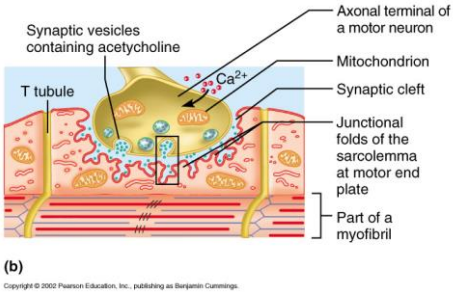




Boron & Boulpaep: Medical Physiology, 2nd Edition.
Copyright © 2009 by Saunders, an imprint of Elsevier, Inc. All rights reserved.

23

Synapses can be fast or slow

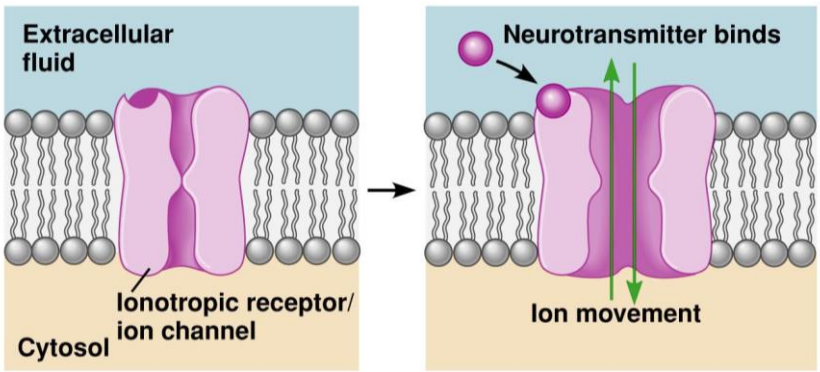


Copyright © 2002 Pearson Education, Inc., publishing as Benjamin Cummings.

24

Two main types of receptors: *i. ionotropic*

Fast response (ionotropic/ligand gated): neurotransmitter binds to a channel linked receptor. Few to several hundred milliseconds: e.g. channels open to allow specific ions through Na^+ , K^+, Cl^-.



(a) Fast response

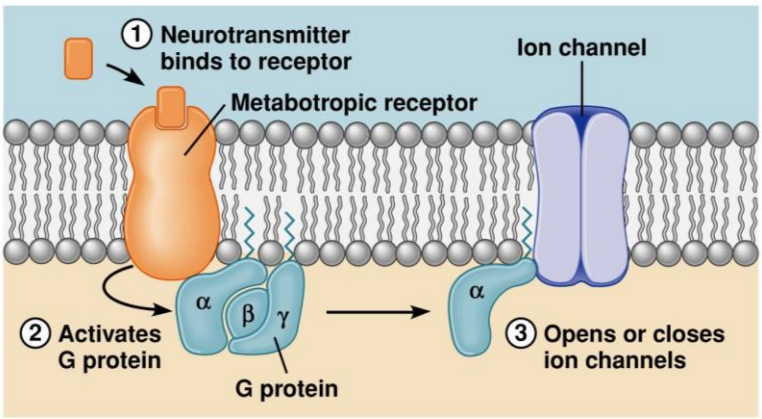
S&G Figure 8.3a

Synaptic transmission

25

Signal Transduction at Synapses*: ii. metabotropic*

– Slow, **G-protein direct coupling to channel**. Metabotropic receptors trigger biochemical changes (metabolic-hence the name) rather than direct changes in membrane permeability.



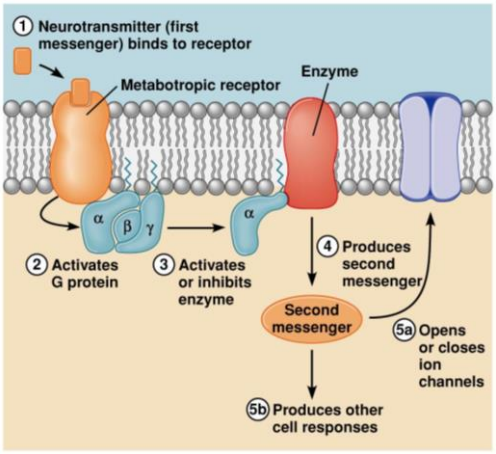
(b) Slow response, direct coupling

Figure 8.3b

26

Signal Transduction at Synapses*: ii. metabotropic*

– Slower, second messenger system: G protein activates/inhibits an enzyme that synthesizes intracellular molecules which then act on specific intracellular targets. **Indirect coupling.**



(c) Slow response, second messenger system

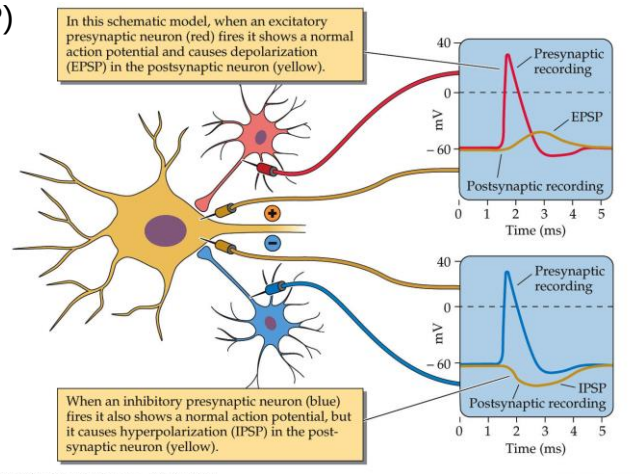
Figure 8.3c

27

Synapses can be excitatory or inhibitory

depolarisation = excitation = excitatory post-synaptic potential (EPSP)

hyperpolarisation = Inhibition = inhibitory post-synaptic potential (IPSP)



Biological Psychology 5e, Figure 3.9

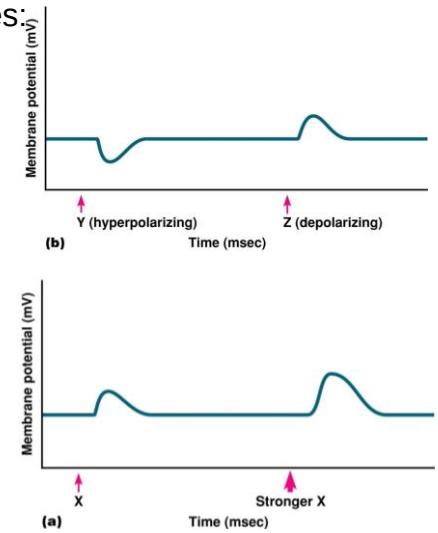
© 2007 Sinauer Associates, Inc.

28

UCL

Different effects of synapses:

- Some are ***depolarizing*** *(EPSP):* some are ***hyperpolarizing*** *(IPSP): depends on the type of transmitter and receptor*
- Magnitude varies depending on size of stimulus



S&G: Figure 7.10 a&b

29

UCL

Excitatory synaptic transmission -
or how neurons excite other neurons

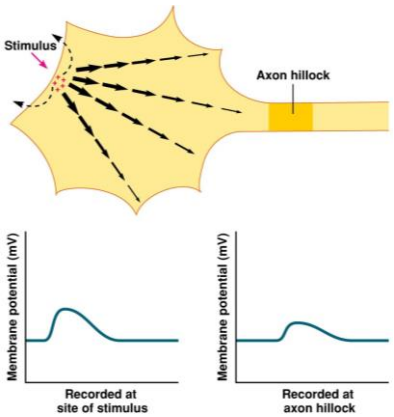
•An excitatory synapse brings the membrane of the post-synaptic neuron closer to threshold for generation of an action potential

•This depolarisation is called an excitatory post-synaptic potential- commonly referred to as EPSP

•EPSPs ***are graded***; the amplitude of depolarisation increases as more transmitter binds to the receptors.

•Depolarisation is greatest at the site of origin and decreases as it moves towards the axon hillock

•EPSPs can be fast or slow

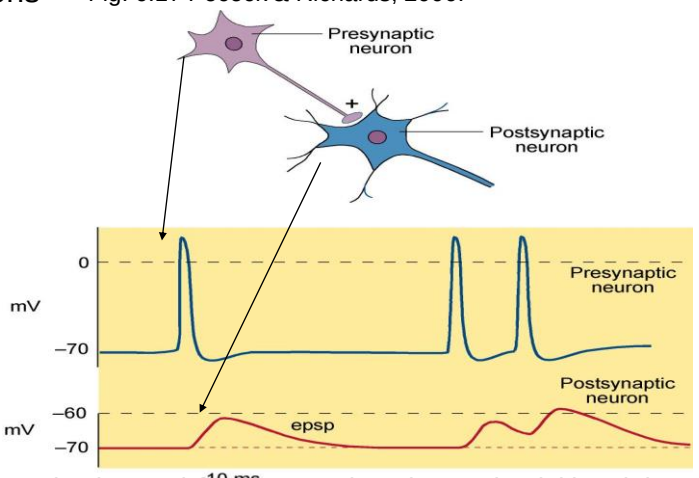


S&G Figure 7.11

30

Excitatory synaptic transmission or how neurons excite other neurons

Fig. 6.17 Pocock & Richards, 2006.

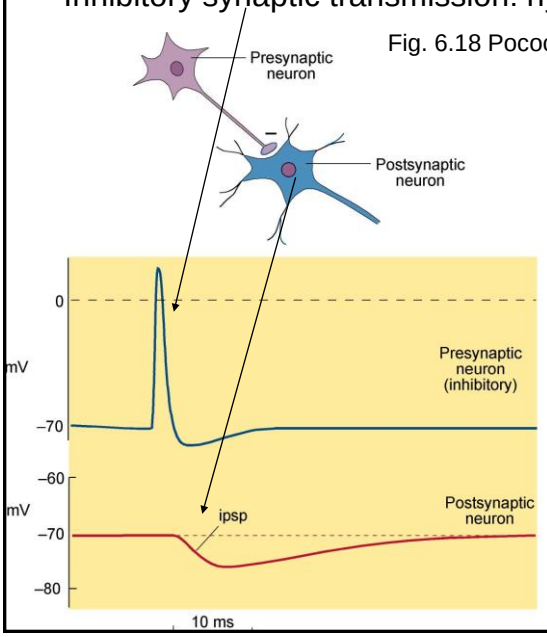


*EPSPs (and IPSPs) **last longer** than the AP that initiated them, they are also **smaller** and can be **superimposed/added** onto one another allowing them to be graded i.e. **summation***

31

Inhibitory synaptic transmission: hyperpolarisation

Fig. 6.18 Pocock & Richards, 2006.

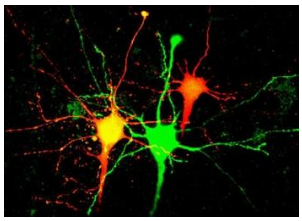


- IPSPs are graded
- Neurotransmitter binds to receptor
- Channels for either K^+ or Cl^- open
- If K^+ channels open
  - K^+ moves out → IPSP
- If Cl^- channels open,
  - Cl^- moves in → IPSP

32

Synaptic potentials

- Are graded in intensity
- May be depolarising (epsps) or hyperpolarising (ipsps)
- Can be fast or slow: slow important for control functions such as cardiac output, secretion of hormones, mood changes etc
- As synaptic potentials are graded, they can summate –unlike action potentials
- Summation is necessary because a single EPSP is rarely sufficiently large to depolarize the postsynaptic neuron to threshold
- Two kinds of summation:
  - Temporal summation
  - Spatial summation



33

Sherrington, on the synapse

- Sir Charles Sherrington (1932)

Nobel prize winner for work on reflex action, threshold, and summation properties of the synapse



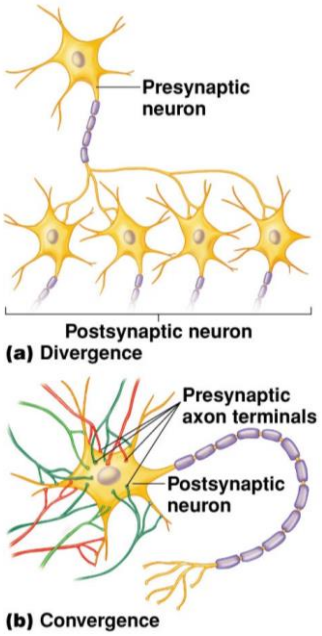
- Spatial summation – balance of all excitatory and inhibitory inputs across dendrites and soma
- Temporal summation – repeated low-level input sums together to cause an output

34

34

UCL

Divergence and Convergence



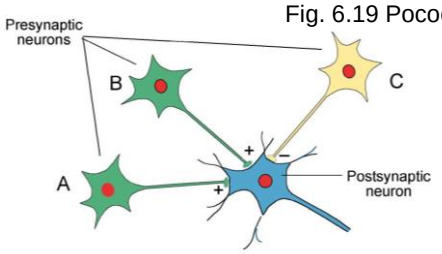
S&G Figure 8.7

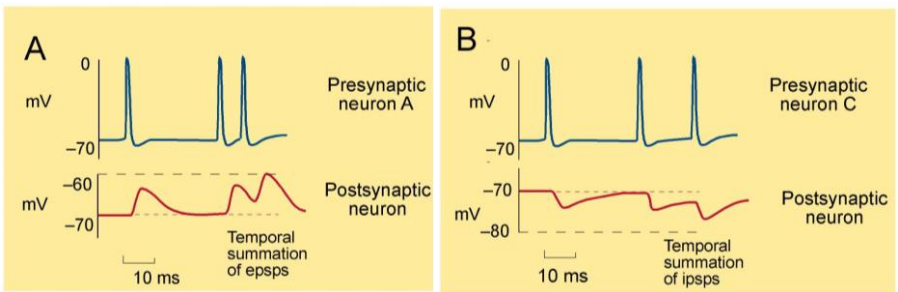
35

UCL

Temporal Summation of synaptic potentials

Fig. 6.19 Pocock & Richards, 2006.

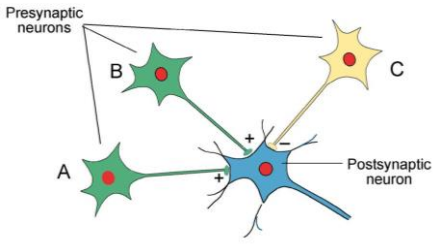


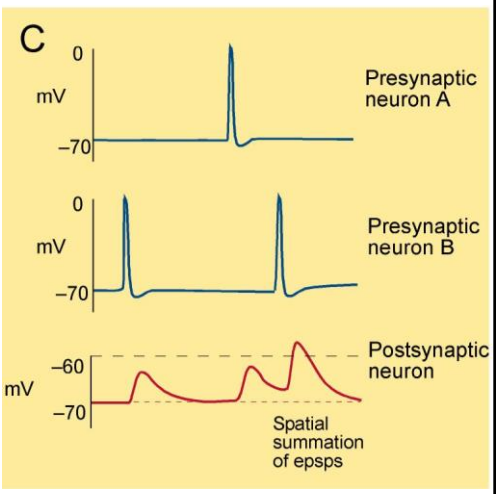


Consecutive arrival of APs

36

Spatial summation of epsps



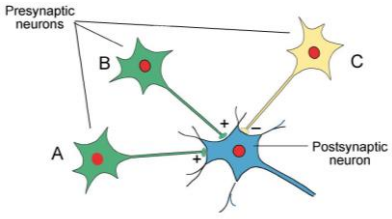


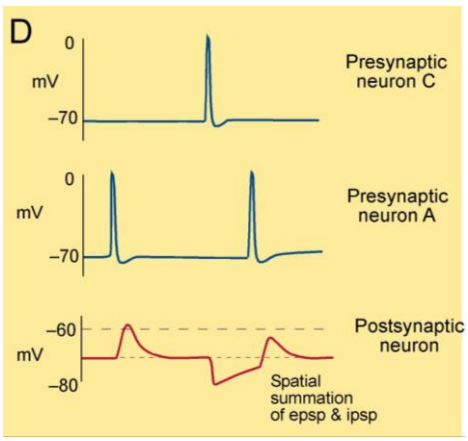
A + B = Depends on nearly ***simultaneous*** timing of occurrence of ***two or more separate stimuli***

Fig. 6.19 Pocock & Richards, 2006.

37

EPSPs and IPSPs summate to regulate the excitability of neurons



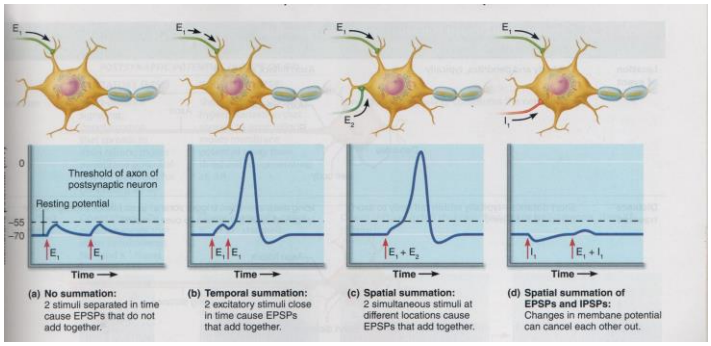


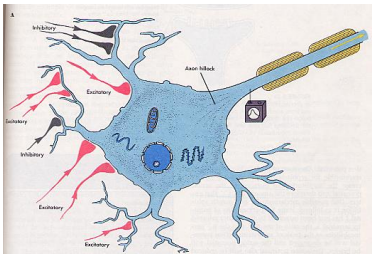
A (epsp) + C (ipsp) = <A

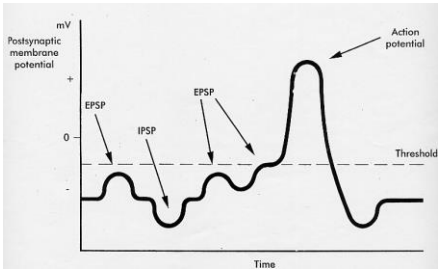
Location and Timing……

Fig. 6.19 Pocock & Richards, 2006.

38

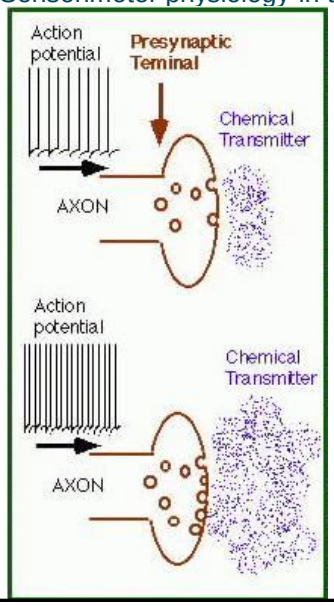






39

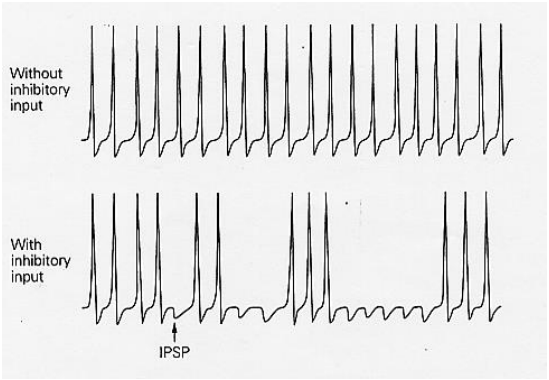
Action potential frequency determines amount of transmitter release [more in lecture about Sensorimotor physiology in term 2]







40



Inhibition shapes excitation to modulate activity

Kandel, Schwartz and Jessell, 2000

41

Synaptic signalling is mediated by a wide variety of substances: ***neurotransmitters*** belong to a variety of different chemical classes

Neurotransmitter		Postsynaptic Effect
	Acetylcholine (ACh)	Excitatory
Amino acids	Glutamate	Excitatory
	γ-Aminobutyric acid (GABA)	Inhibitory
	Glycine	Inhibitory
Biogenic amines	Dopamine	Excitatory (via D1 receptors)
		Inhibitory (via D2 receptors)
	Norepinephrine	Excitatory
	Epinephrine	Excitatory
	Serotonin	Excitatory or inhibitory
	Histamine	Excitatory
Purines	Adenosine triphosphate (ATP)	Excitatory/ neuromodulatory
Neuropeptides	Substance P	Excitatory
	Metenkephalin	Inhibitory
	Opioids	Inhibitory
	Adrenocorticotropin	Excitatory

42

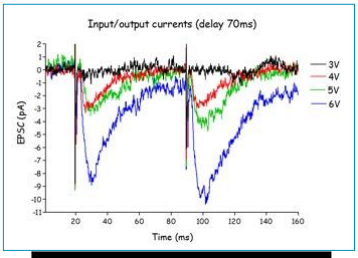
Property	Action potential	Synaptic (graded) potential
Amplitude	Large (70-110mV)	Small, but can summate to result in AP
Duration	Brief: usually 1-2 msec	Variable: 5msec to minutes
Conduction	All or none	Decremental
Summation	None, amplitude is same regardless of stimulus initiating it	Graded: depends on condition initiating it
Signal	Depolarizing	Depolarizing or hyperpolarizing
Refractory period	Absolute and relative	None
Distance	Long	Short

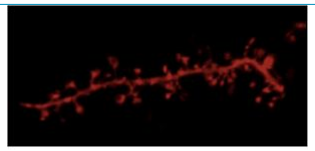
43

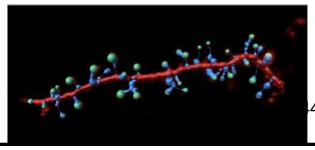
UCL research focus!!!!!!!!!!

https://www.ucl.ac.uk/biosciences/departments/npp/people/iris-profiles/edwards-frances

"A lot of past work has focused on such homeostasis but generally by studying the effects of weakening or strengthening all the synapses of the neurones measured, using pharmacological means.
Instead we directly visualise changes in the synapses (De Simoni et al., 2006) as specialising in high resolution electrical recording between individual neurones or recording of plasticity in the network."
"Memory must involve activity-dependent changes in the network of communication between brain cells.
The hippocampus has long been known to be involved in the laying down of memory and much work on this field has concentrated on this area of the brain."

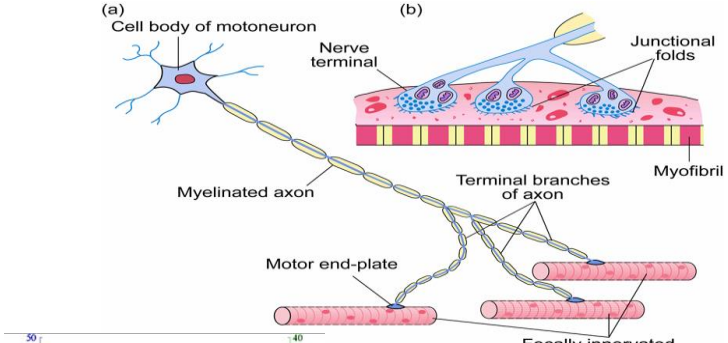


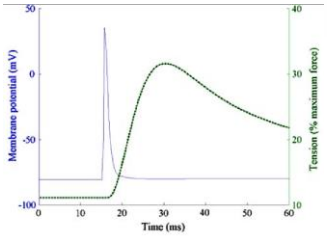


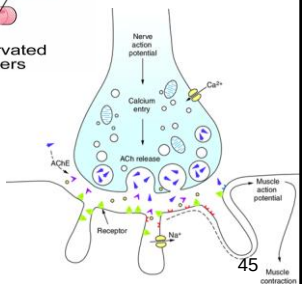


44

Specific example of fast transmission = Neuromuscular junction: motoneurones control activity of skeletal muscle fibres (the motor unit)







45

Neuromuscular transmission: fast synaptic signalling

•AP originate in cell body and propagate along axon and depolarises nerve terminal-

•Signalling molecule released- calcium- and calcium ions flow into the nerve terminal down their electrochemical gradient. Local rise in Ca^{2+} results in fusion of docked synaptic vesicles with plasma membrane.

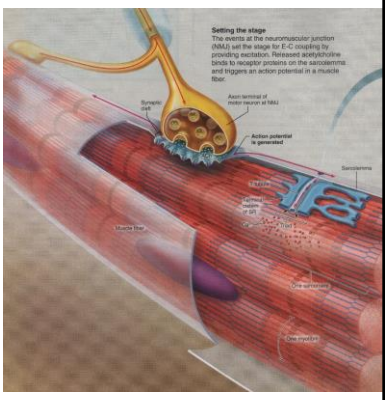
•ACh released into synaptic cleft

•ACh diffuses across and binds with **nicotinic** receptors on post-synaptic membrane

•Ion channels open and flow of ions cause [Na^+/K^+] depolarisation muscle membrane in end-plate region (end plate potential; epp)

•When epp at threshold AP generated and propagated down muscle fibre. ACh broken down and recycled.

•Muscle AP causes release of Ca^{2+} from sarcoplasmic reticulum and triggers contraction of muscle.



46

Neuromuscular transmission: fast synaptic signalling

•AP originate in cell body and propagate along axon and depolarises nerve terminal-

•Signalling molecule released- calcium- and calcium ions flow into the nerve terminal down their electrochemical gradient. Local rise in Ca^{2+} results in fusion of docked synaptic vesicles with plasma membrane.

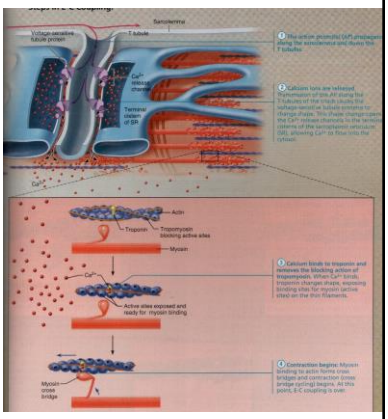
•ACh released into synaptic cleft

•ACh diffuses across and binds with **nicotinic** receptors on post-synaptic membrane

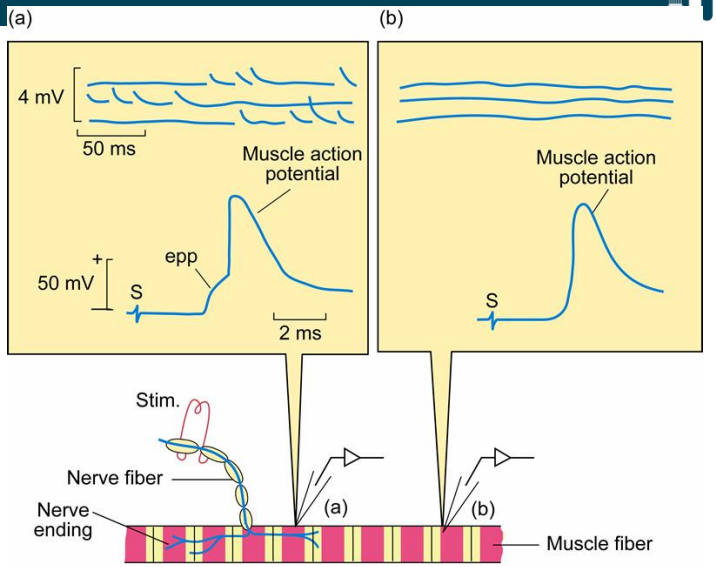
•Ion channels open and flow of ions cause depolarisation muscle membrane in end-plate region (end plate potential; epp)

•When epp at threshold AP generated and propagated down muscle fibre. ACh broken down and recycled.

•Muscle AP causes release of Ca^{2+} from sarcoplasmic reticulum and triggers contraction of muscle [interaction of actin/myosin].



47



Epp's (depolarisation of muscle membrane at motor end-plate) and mepp's (miniature epps; random and infrequent)- only seen in end-plate region

48

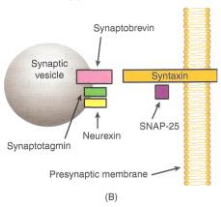
Synapses can be influenced by various kinds of agents





Pre-synaptic: prevent release of ACh.

Botulinum toxin: attacks one of the fusion proteins (SNAP-25, syntaxin or synaptobrevin) at the neuromuscular junction, preventing vesicles from anchoring to the membrane to release acetylcholine.

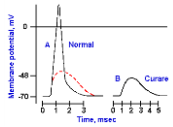


49

Synapses can be influenced by various kinds of agents

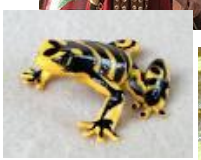
Strychnos toxifera





Post-synaptic: prevent uptake of ACh by the nAChR.





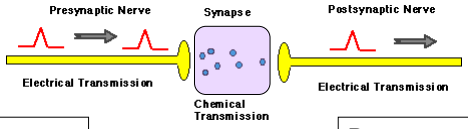


Post-synaptic: Na^+ channels in muscle cells remain open. BactrachoTX acts on the open channel and stops it closing. (also prevents APs in axons)



50

Synaptic modulation: underlies learning and memory and recovery processes.



Presynaptic:

Modification of the amount of transmitter release

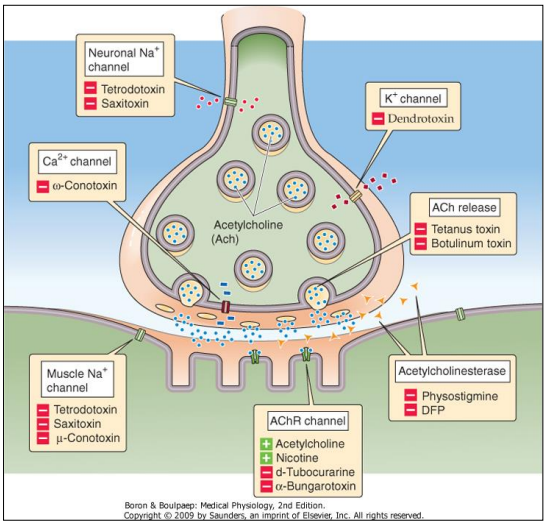
Synaptic cleft: modify reuptake of transmitter

Post-synaptic:

Greater sensitivity to the transmitter: modify receptor/ion channel

51

Multiple ways in which synapses can be influenced pharmacologically



52

Summary of synaptic transmission 2

Synapses are the way that cells communicate with each other.

It is a 'one-way' signalling mechanism

Synapses can be of different types, mechanism and effect.

Synaptic potentials are graded unlike the action potential which is 'all or none'.

They last longer than the action potential that initiated them and therefore they can summate: temporal and spatial summation

Synaptic transmission can be slow or fast depending on its function.

Watch: synapses

The most studied synapse is the neuromuscular junction- this is an example of fast synaptic signalling which underlies most of our control of voluntary movement.

Synapses can be modulated: basis of neuroplasticity (see slide 46).

Information is coded by the sequence of arrival of action potentials

53